

Developmental amygdala scRNA-seq

E18 · P0 · P10 · P21 — an exploratory update for Wei-Ming Kao

What changes when the early libraries are added?

76,446 putative singlets
10 libraries across four stages
25,029 shared genes

Population inventory
P10 transcription factors
Guidance / remodelling candidates

This dataset can generate testable candidates. It cannot explain a MEF2C-KO projection phenotype by itself.

What is in the combined dataset?

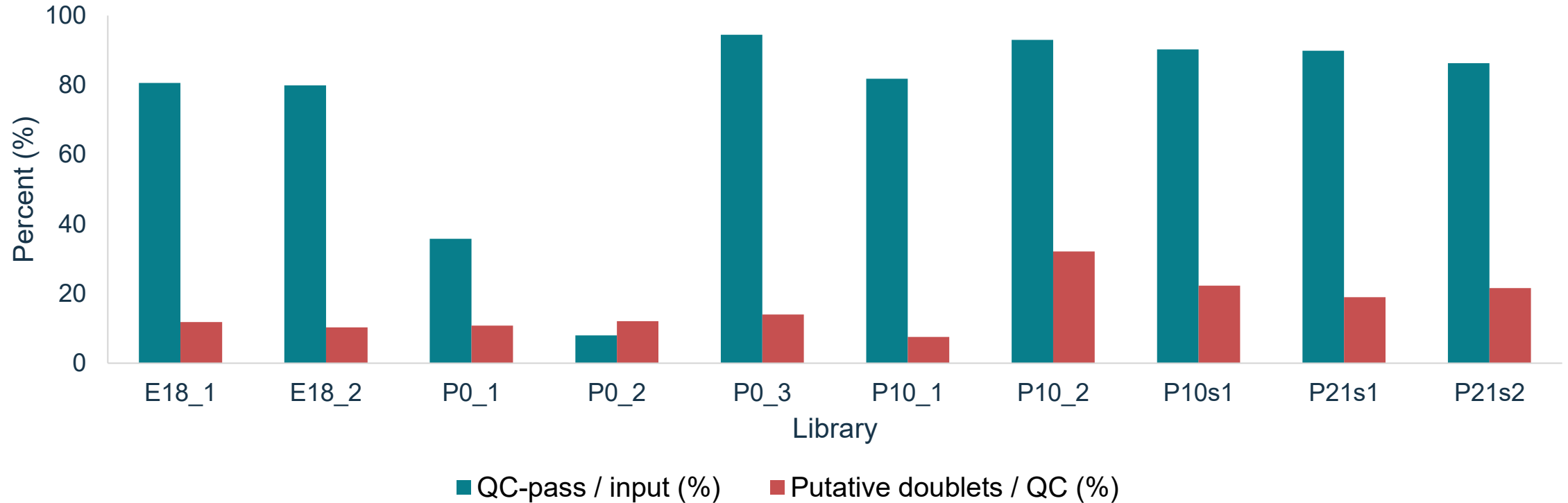
Seven new libraries + three previous P10/P21 libraries

Library	Stage	Source	Input	QC-pass	Putative singlets
E18_1	E18	New	4,686	3,779	3,332
E18_2	E18	New	5,026	4,018	3,604
P0_1	P0	New	21,750	7,784	6,943
P0_2	P0	New	60,941	4,888	4,296
P0_3	P0	New	4,872	4,604	3,960
P10_1	P10	New	40,994	33,548	31,017
P10_2	P10	New	18,458	17,171	11,654
P10s1	P10	Previous	7,254	6,546	5,086
P21s1	P21	Previous	3,141	2,824	2,287
P21s2	P21	Previous	6,309	5,443	4,267

Shared-gene analysis avoids treating the 7,256 new-only genes as absent in old samples.

QC is a major source of uncertainty

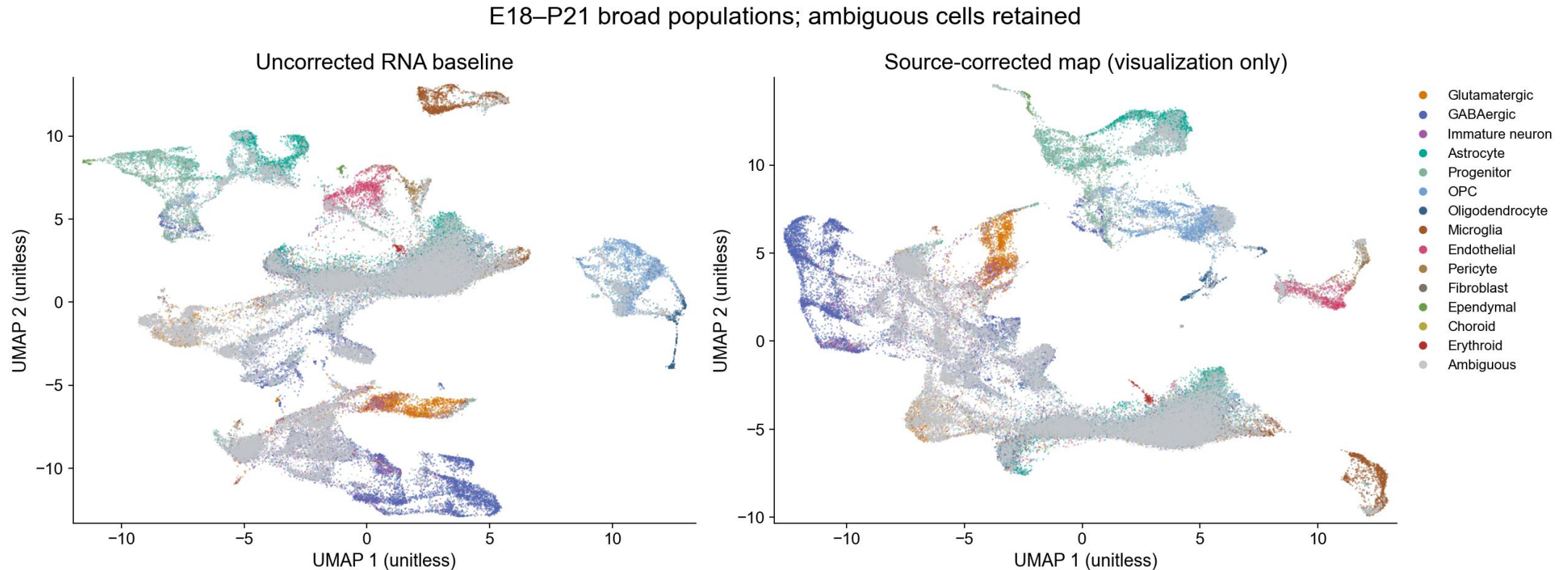
Simple QC retention and classifier-labelled doublets — separate measures



P0_2: only 8.0% pass QC. P10_2: 32.1% putative doublets. Old/new mitochondrial capture differs sharply.

A broad map, not a cell-type verdict

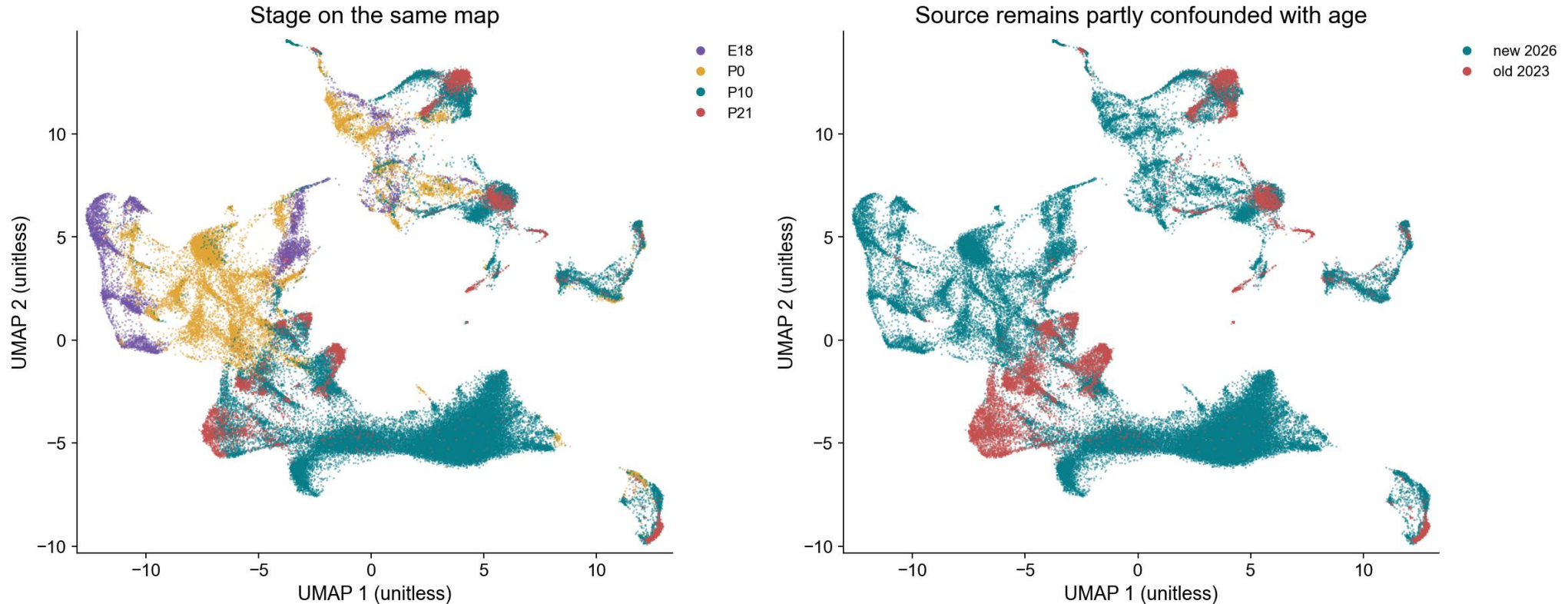
Both uncorrected and source-corrected UMAPs are preserved



Labels use excluded markers and reference checks. Ambiguous cells remain visible; fine developmental identities are provisional.

Stage and source are partly confounded

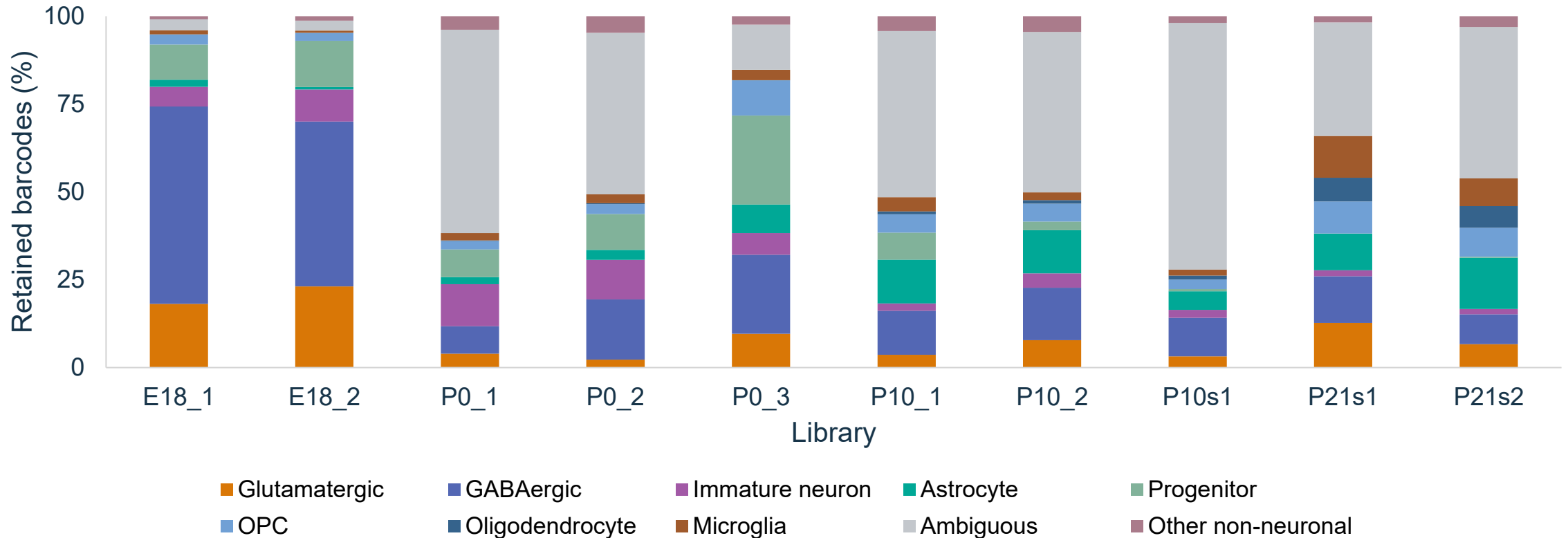
P10 overlaps sources; P21 exists only in the previous source



Harmony is for visualization only. All expression calculations use unchanged, uncorrected RNA counts.

Captured populations differ by library

Fractions of retained barcodes — not proportions in intact tissue



Library capture, depth, ambiguous labels and doublet selection prevent a definitive developmental cell census.

Annotation checks expose unresolved cells

The adult-atlas-only baseline did not validate well

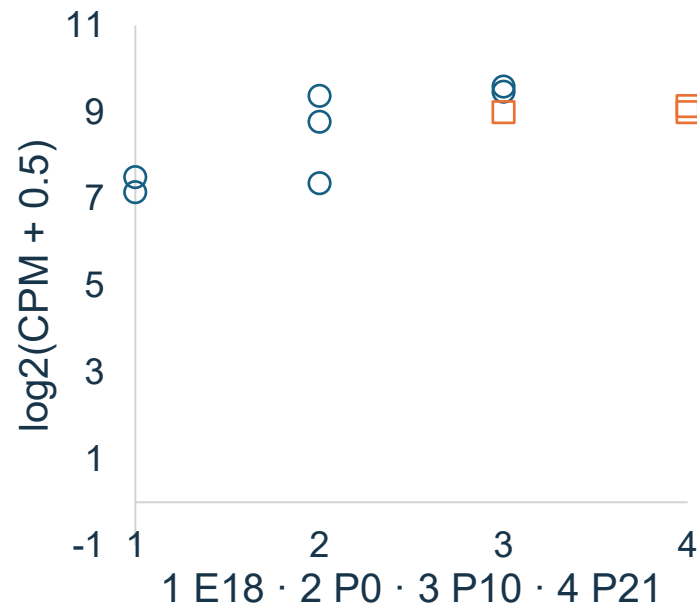
Library	Adult atlas / marker	Previous broad split	Ambiguous (QC %)
E18_1	6.6%	75.0%	2.8%
E18_2	3.8%	74.7%	2.5%
P0_1	10.9%	67.3%	55.0%
P0_2	15.6%	72.3%	44.0%
P0_3	22.3%	84.3%	11.9%
P10_1	25.8%	59.1%	45.1%
P10_2	25.8%	60.2%	39.5%
P10s1	35.4%	94.7%	67.1%
P21s1	58.3%	97.2%	30.0%
P21s2	56.2%	95.9%	40.7%

Broad marker-supported assignments are useful candidates. Novel or low-depth identities require further validation.

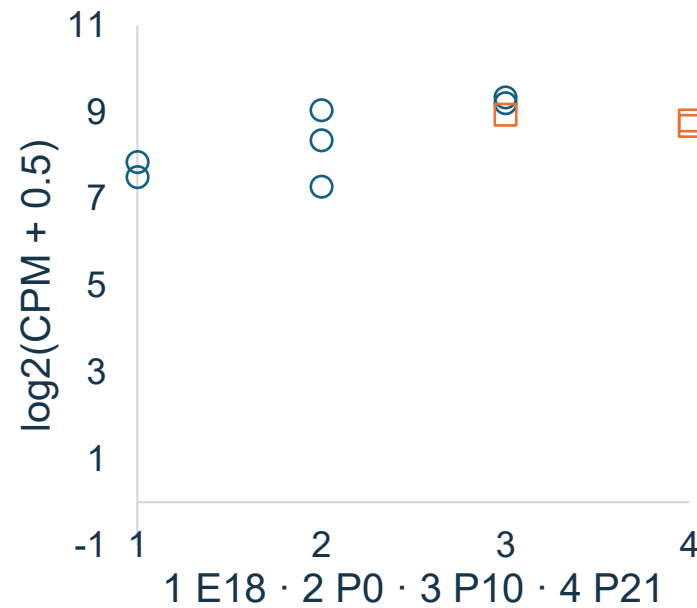
Mef2c expression: library-level evidence

Each dot is one library; old and new P10 are shown separately

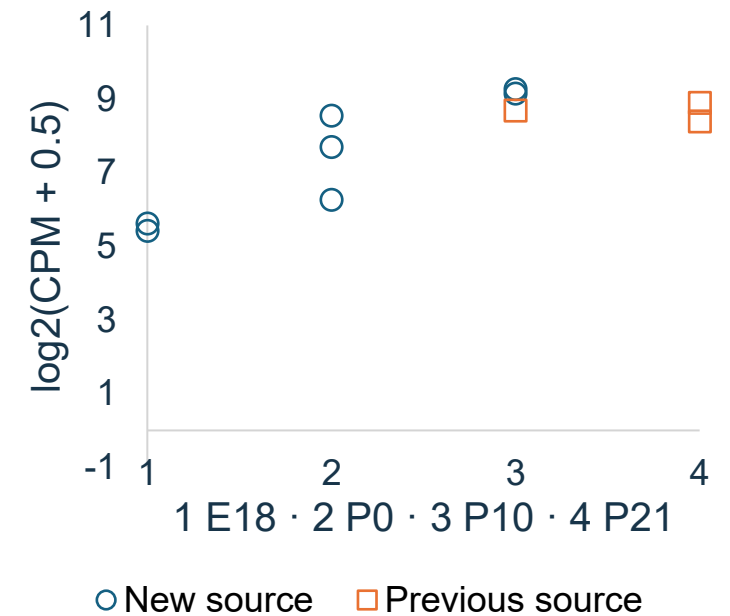
Glutamatergic



GABAergic



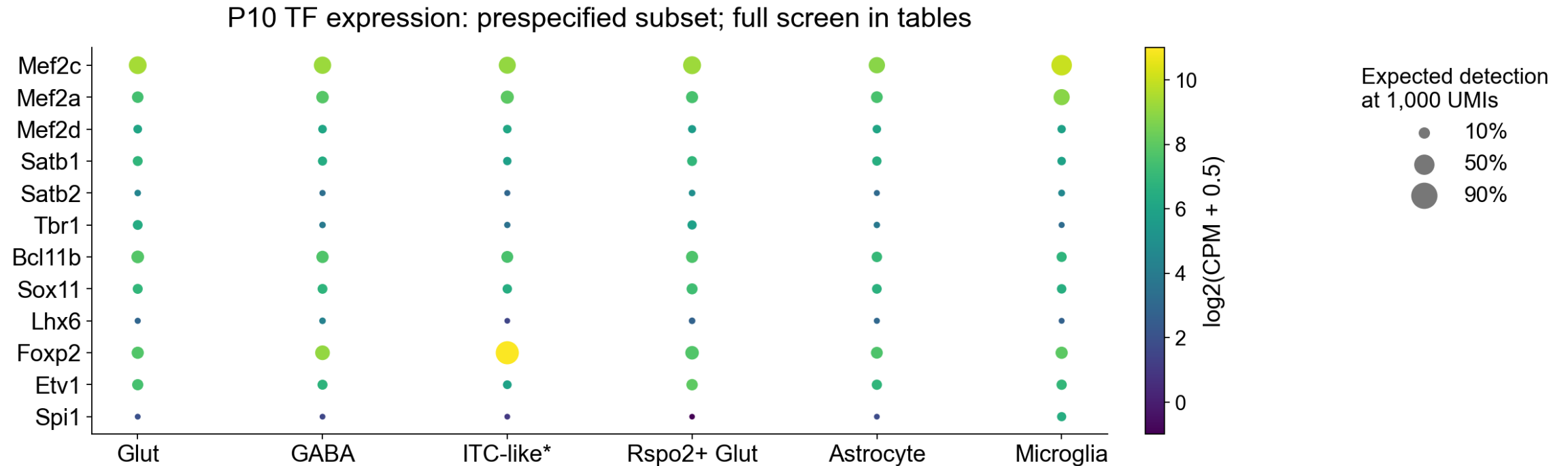
ITC-like GABA*



*ITC-like identity is provisional and not spatially verified. Similar expression does not imply similar MEF2C function.

P10 transcription-factor expression

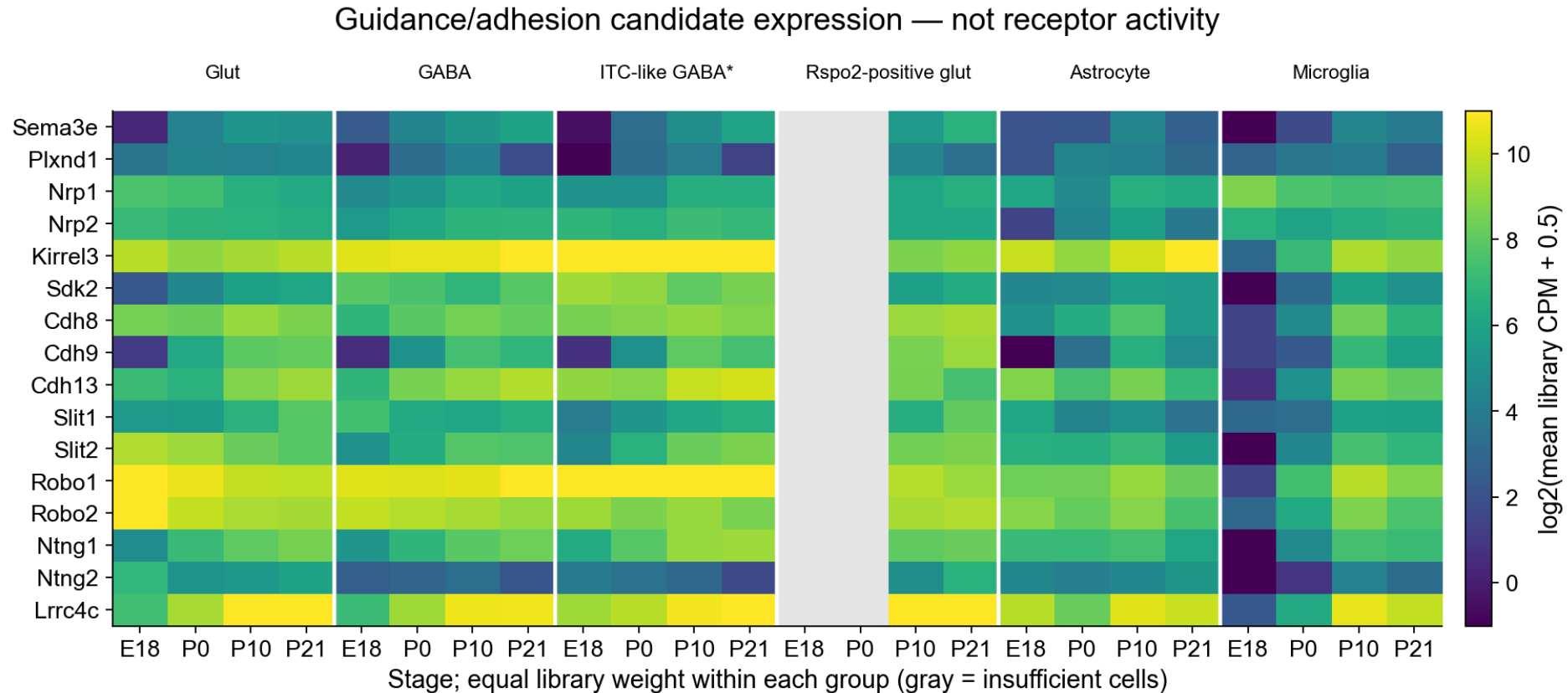
A candidate screen, not TF activity or a causal target ranking



Use cell identity + candidate availability to plan experiments; verify protein, location and genotype-specific regulation.

Guidance and adhesion candidates across stages

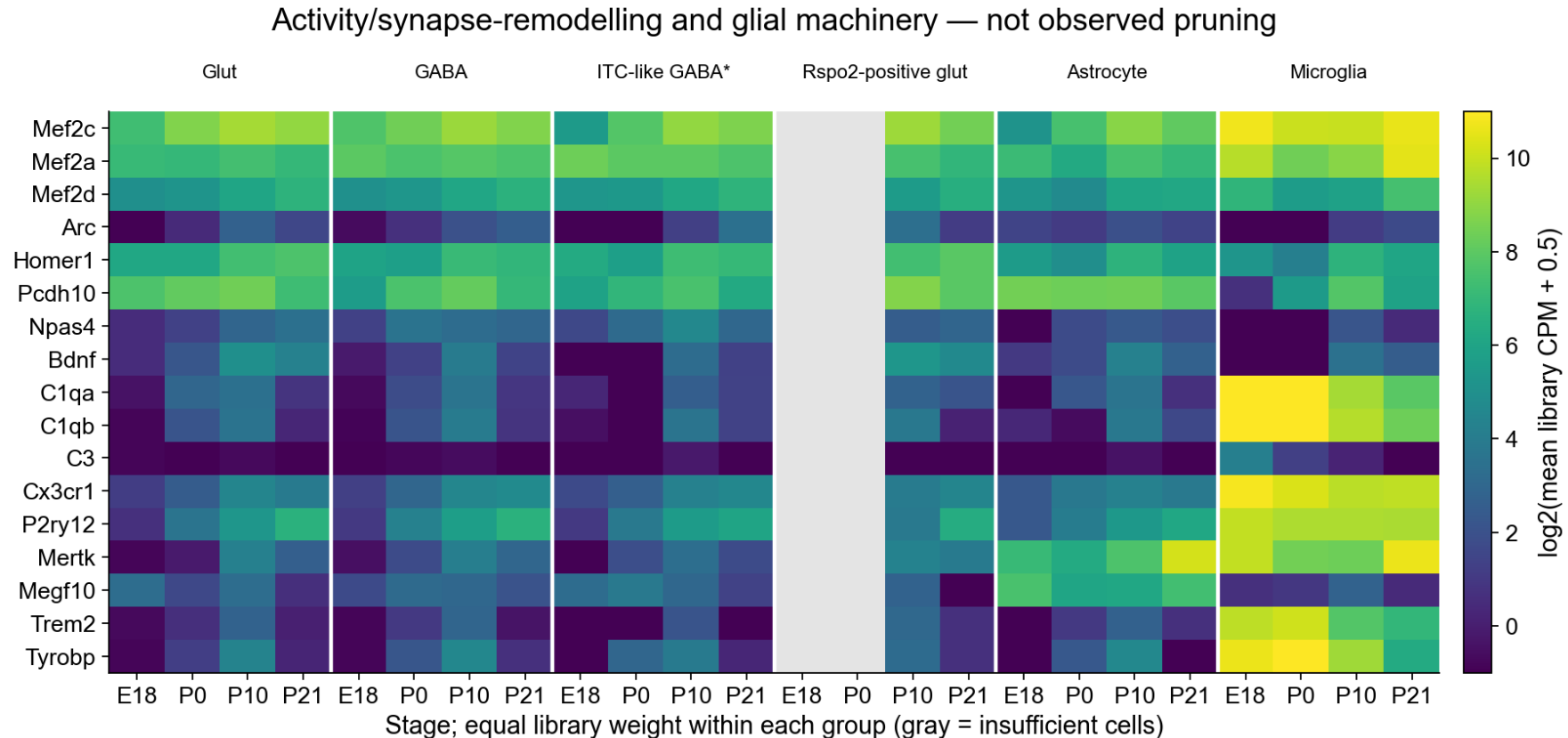
Sema3e · Plxnd1/Nrp1 · Kirrel3 · cadherins · Slit/Robo



Expression nominates components. Functional cue response and the projecting cortical cell must be tested independently.

Synapse-remodelling and glial candidates

MEF2-associated activity genes and engulfment/complement machinery



These observations do not distinguish retained axon branches, additional synapses, altered tracer labeling or changed cortical cell identity.

Cortical context is external adult reference

The new experiment samples amygdala, not the projecting cortex

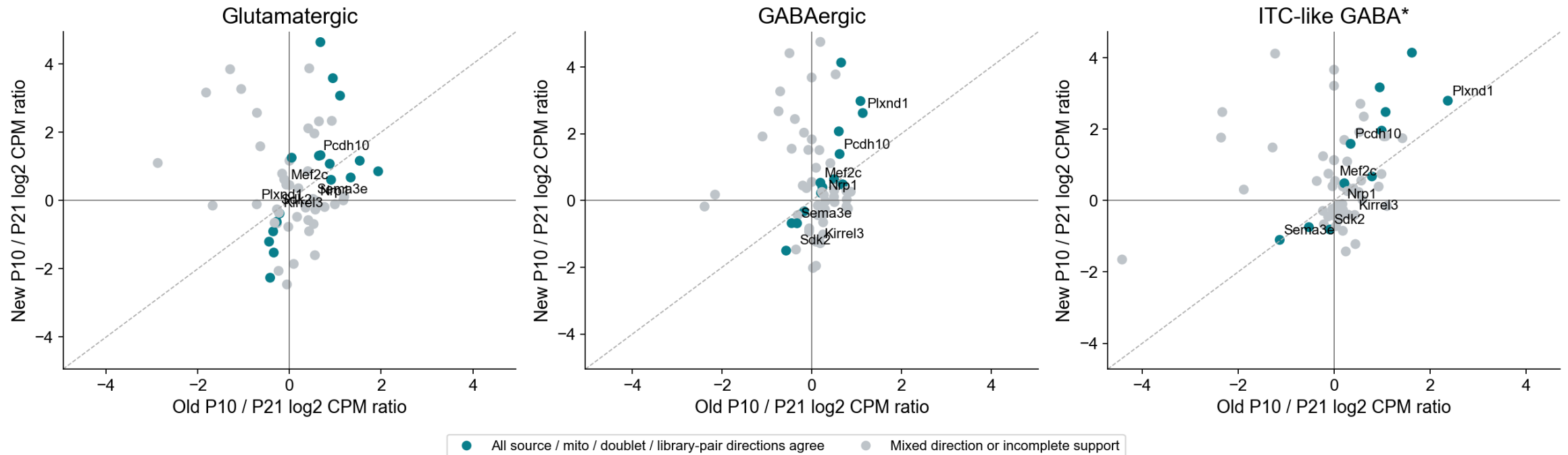
Gene	L2/3 IT Detection %	L4/5 IT Detection %	L5 ET Detection %
Kirrel3	89.9	73.0	99.1
Plxnd1	24.8	50.0	7.1
Nrp1	45.7	34.0	75.0
Nrp2	not available	not available	not available
Sdk2	52.1	44.6	91.0
Robo1	98.6	91.4	99.9

Do not claim Plxnd1 is exclusive to one cortical class, or that Nrp1-negative RNA means a repulsive receptor state.

P10–P21 directions must survive source checks

True pseudobulk CPM ratios; mito and putative-doublet sensitivity

Developmental direction must survive source and QC checks



A consistent direction is stronger descriptive evidence, but P21-only-old-source confounding and unknown animal n remain.

Core hypotheses: consistent and unstable patterns

Prespecified genes shown whether they pass or fail the sensitivity rule

Gene / population	Old P10 / P21 log2 ratio	New P10 / P21 log2 ratio	All direction checks
Mef2c — Glut	-0.07	0.46	Mixed / unstable
Mef2c — GABA	0.19	0.53	Consistent direction
Pcdh10 — Glut	0.64	1.31	Consistent direction
Kirrel3 — ITC-like GABA*	0.43	-0.41	Mixed / unstable
Sema3e — Glut	0.52	0.04	Mixed / unstable
Plxnd1 — ITC-like GABA*	2.37	2.79	Consistent direction
Nrp1 — Glut	0.57	-0.01	Mixed / unstable

Pcdh10 is direction-consistent. Kirrel3 ITC-like change reverses by source. Neither observation is a KO result.

Three scenarios remain plausible — not established

More cortical labeling in BLA can have several explanations

Scenario	Candidate components	Decisive independent test
Retained branches / synapses	MEF2 activity pathway, Pcdh10; glial engulfment machinery	Sparse axons at several ages + boutons/synapses, normalized to cortical labeling
Altered cue response / matching	Sema3e–Plxnd1/Nrp1, Kirrel3, cadherins, Slit/Robo	KO/WT projecting cortical neurons; protein/cue-response and rescue tests
Cell identity / labeling difference	Cortical fate, survival, reporter expression or tracer uptake	Count labeled cortical neurons; match layer/type, injection and survival

BLA RNA availability helps choose a test. It cannot decide which scenario caused the projection phenotype.

A practical follow-up for the wiring project

Prioritize measurements that distinguish mechanisms

1. Establish the anatomical phenotype

Same-age KO/WT; normalize axon signal to cortical labeling and injection.
Separate axon branches, boutons, synapses and reporter intensity.

2. Identify both sides of the connection

Spatially validate P10 candidate BLA / ITC-like cells; profile or label the actual cortical projection neurons rather than using adult atlas alone.

3. Test one candidate mechanism with a rescue

Choose guidance/matching versus branch retention only after these checks.

Reproducible pipeline and data handoff

Original inputs are preserved; all important decisions are documented

Step	What it does	Saved evidence
00 / 01	Input audit → shared-gene QC → putative doublets	Manifest, QC, barcode check, classifier calls
02 / 03	Marker/reference checks → two maps	Predictions, ambiguous labels, RNA integrity
04 / 05	Library pseudobulk → sensitivity → figures	CPM, detection, TF/candidate tables
PowerPoint	Editable charts/tables + scientific plots	Builder, source JSON, rendered/validated deck

Start: read analysis_development_v2/README.md. Load the processed object with readRDS(); use RNA counts for reanalysis.

Bottom line for collaboration

A useful developmental resource — with explicit uncertainty

Available now

Four-stage maps and provisional populations
P10 TF and candidate-gene expression
Per-library P10/P21 source/QC checks
Reproducible R/Python pipeline

Not established

MEF2C KO transcriptional changes
Direct MEF2 targets or TF activity
Cortical axon pruning / cue switching
Definitive developmental tissue fractions

The strongest next step is matched KO/WT projection-neuron data plus anatomical validation, not a causal claim from UMAP.